Name:Irisha Deshpande **Class:T1** **Roll No:22**

EXPERIMENT:8

Aim: To implement CI/CD pipeline to host a website on AWS

Lab Outcome: LO3

LO3: Build and manage automated CI/CD pipelines for application development, testing, packaging, and deployment.

Theory:

Continuous Integration and Continuous Deployment (CI/CD)

Continuous Integration (CI) is a software development practice where developers merge their code changes frequently into a central shared repository, followed by automated builds and automated test executions to detect integration bugs early. Continuous Delivery/Deployment (CD) extends this practice by automating the release process so that verified code changes can be automatically packaged and deployed directly to staging or production environments without manual intervention.

Implementing CI/CD drastically shortens the software development lifecycle (SDLC), minimizes manual deployment errors, improves software stability, and enables rapid iteration of software products.

Key Components of AWS CI/CD Architecture

- Source Repository (GitHub): Acts as the centralized version control system (VCS). Webhooks or OAuth connection points alert the cloud pipeline instantaneously whenever commits are pushed to specific branches (e.g., main).
- AWS CodePipeline: A continuous delivery orchestration service that models, visualizes, and coordinates every step of the release process. It manages the

stages—such as Source, Build, and Deploy—moving build artifacts sequentially through execution pipelines.

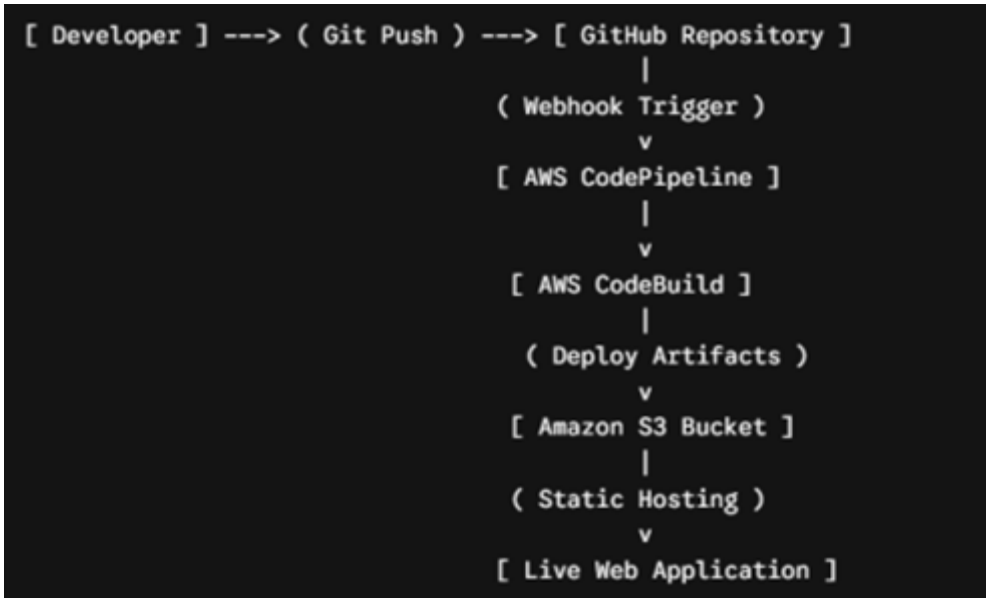
- AWS CodeBuild: A fully managed, scalable build and compilation service. CodeBuild spins up isolated ephemeral compute containers, runs configuration scripts specified in a buildspec.yml file, runs unit tests, compiles static assets, and bundles output artifacts for downstream deployment.
- Amazon Simple Storage Service (Amazon S3): Provides scalable object storage with built-in Static Website Hosting. When configured with public read access (s3:GetObject) and an index.html root document, it serves as a lightweight, highly durable, serverless web server directly from bucket storage.
- AWS Identity and Access Management (IAM): Enforces fine-grained least-privilege security. Service roles ensure CodeBuild and CodePipeline have the exact permissions required (such as read/write access to S3 artifact and deployment buckets) to operate securely without exposing root or persistent user credentials.

Operational Workflow

- Push: A developer commits changes to the GitHub repository.
- Trigger: GitHub triggers an automated event notification to AWS CodePipeline via webhook integration.
- Build: CodePipeline retrieves the source files and triggers AWS CodeBuild to execute containerized compilation and packaging.
- Deploy: The pipeline extracts the verified build artifacts directly into the designated S3 deployment bucket.
- Serve: The Amazon S3 static website hosting endpoint renders the live web application immediately to users.

Architecture & Workflow:

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Implementation:

Phase 1: AWS CodeBuild Project Configuration

Step 1: Create a Build Project

1. Open the AWS Management Console and navigate to Developer Tools > CodeBuild.
2. Click on Create build project.
3. Under Project configuration, set the Project name to pwa-codebuild-vb.
4. Select Default project as the project type.

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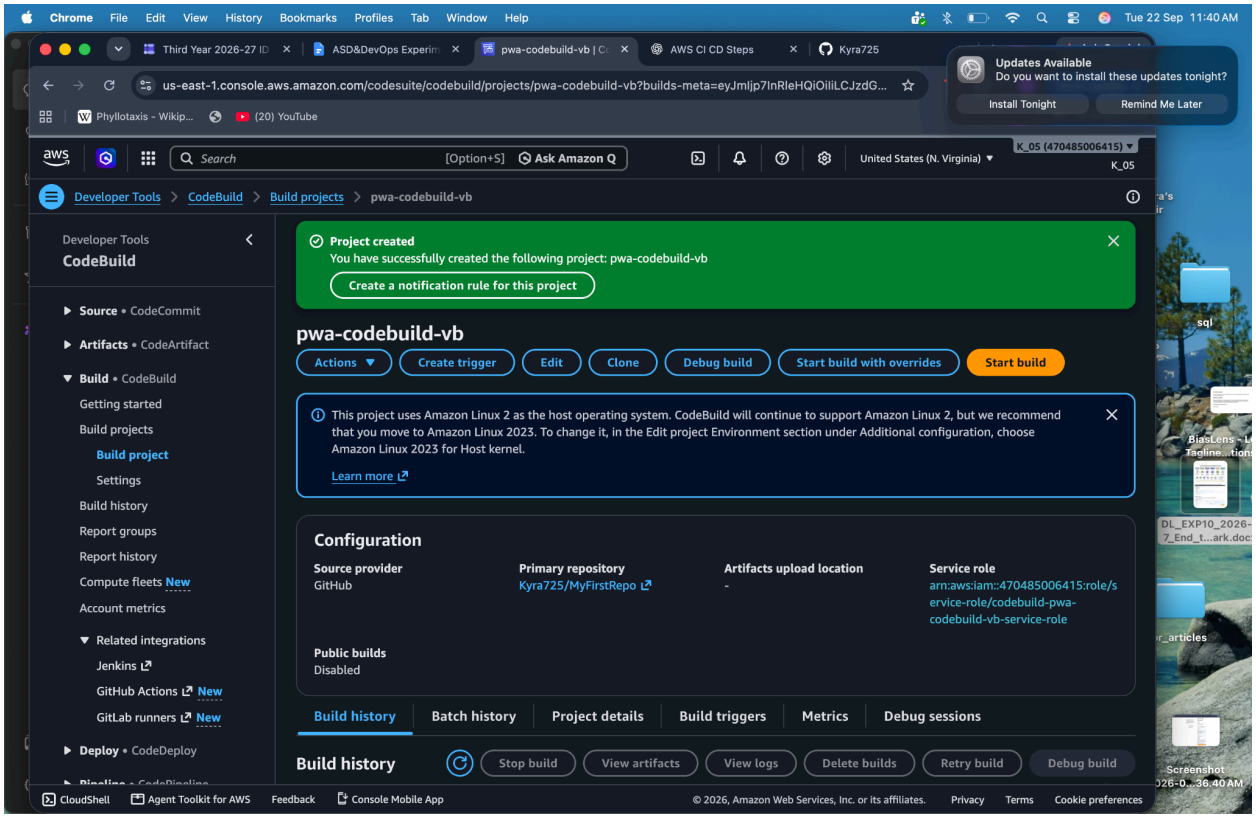
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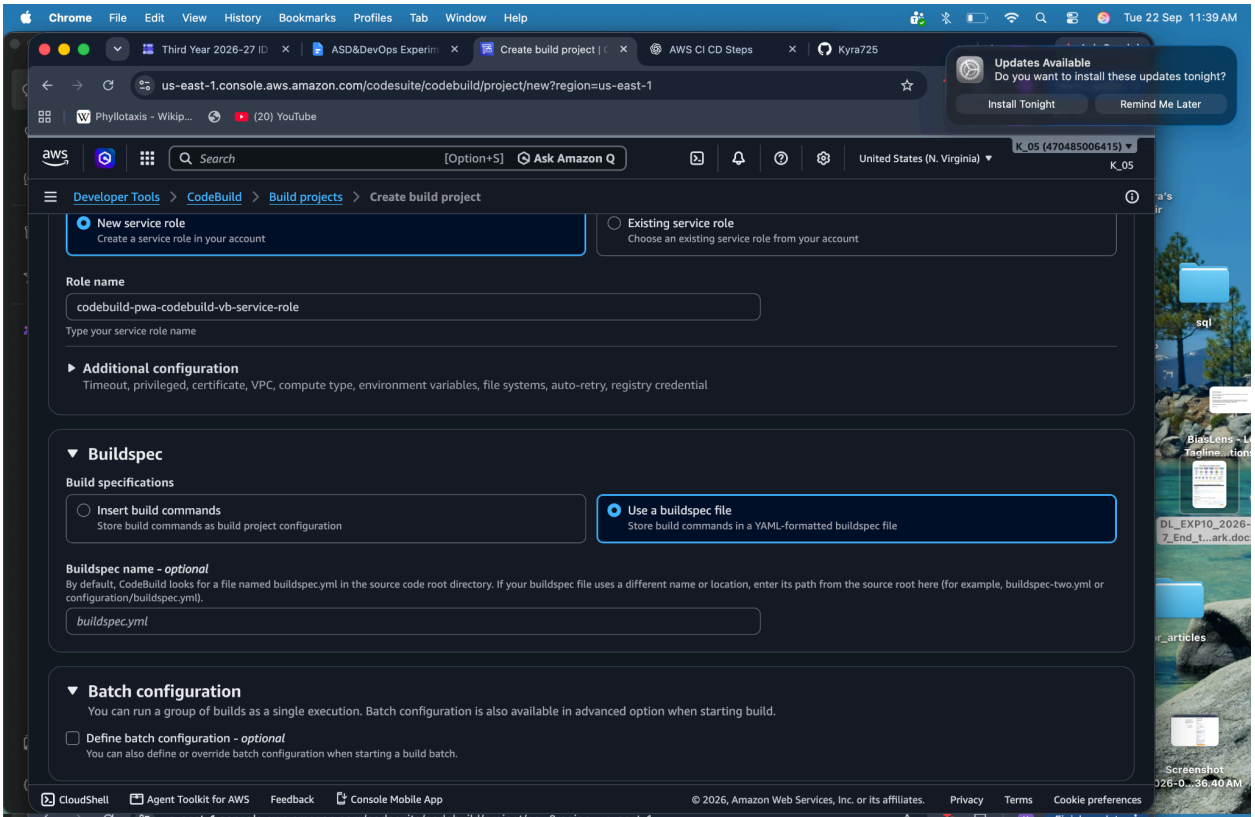
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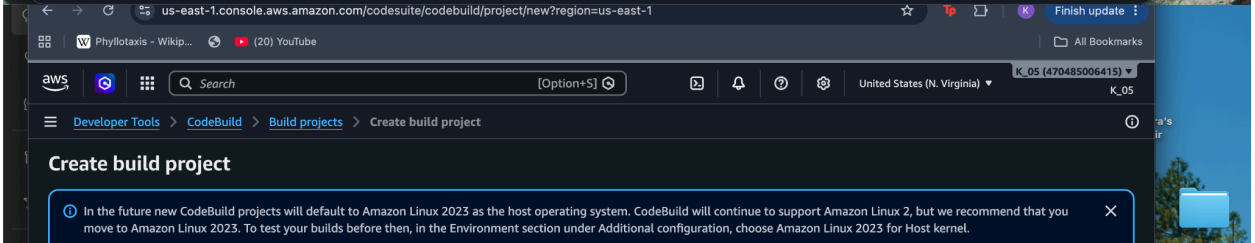
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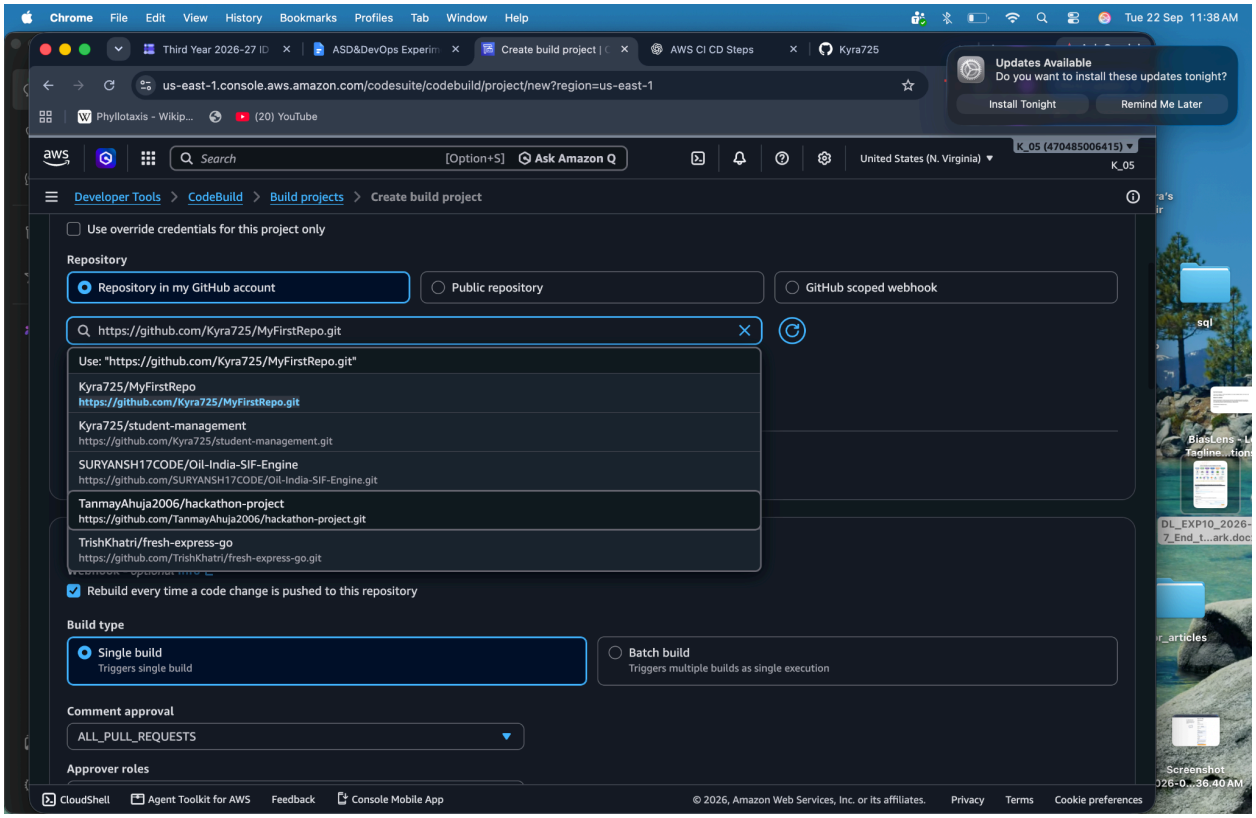
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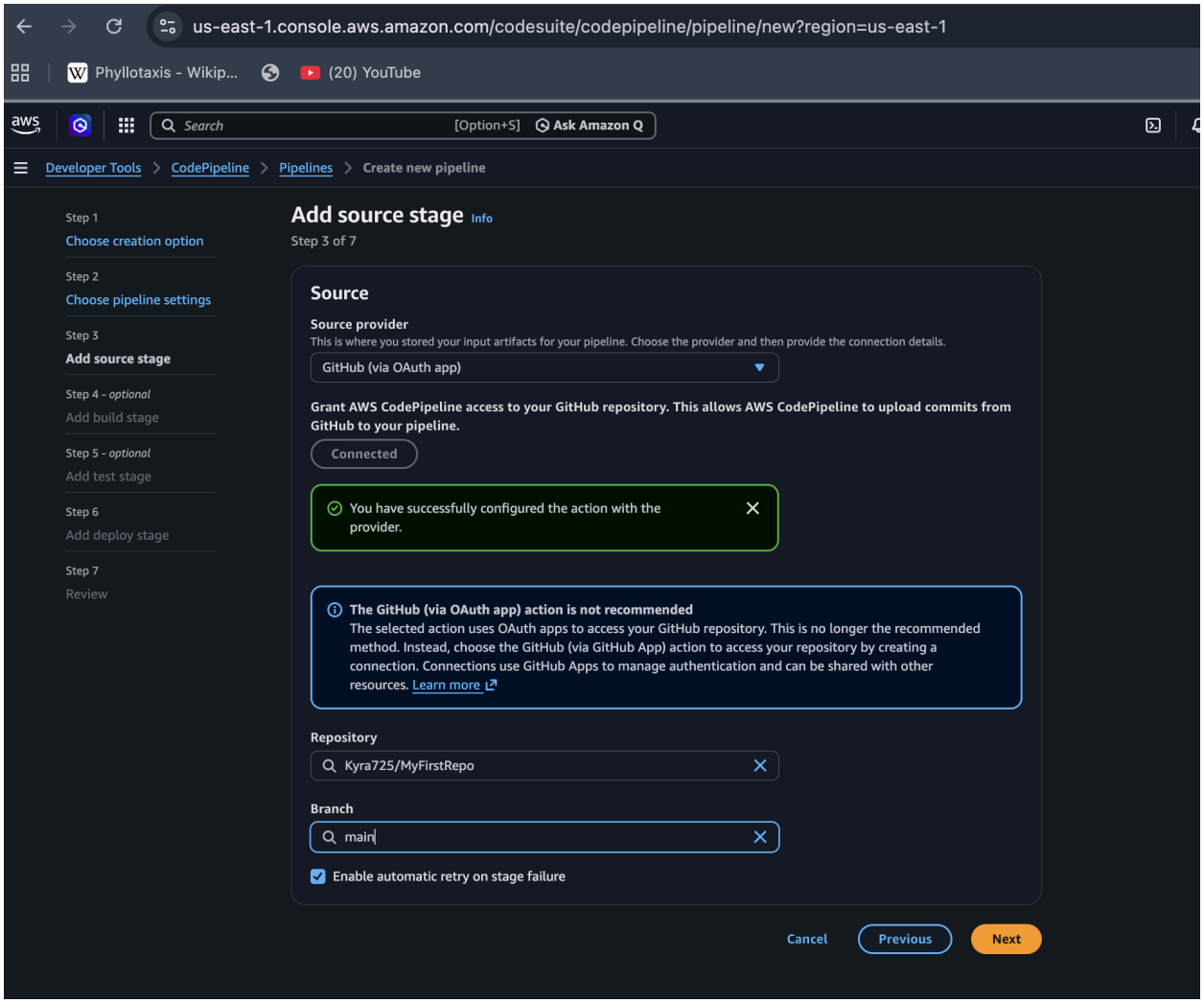
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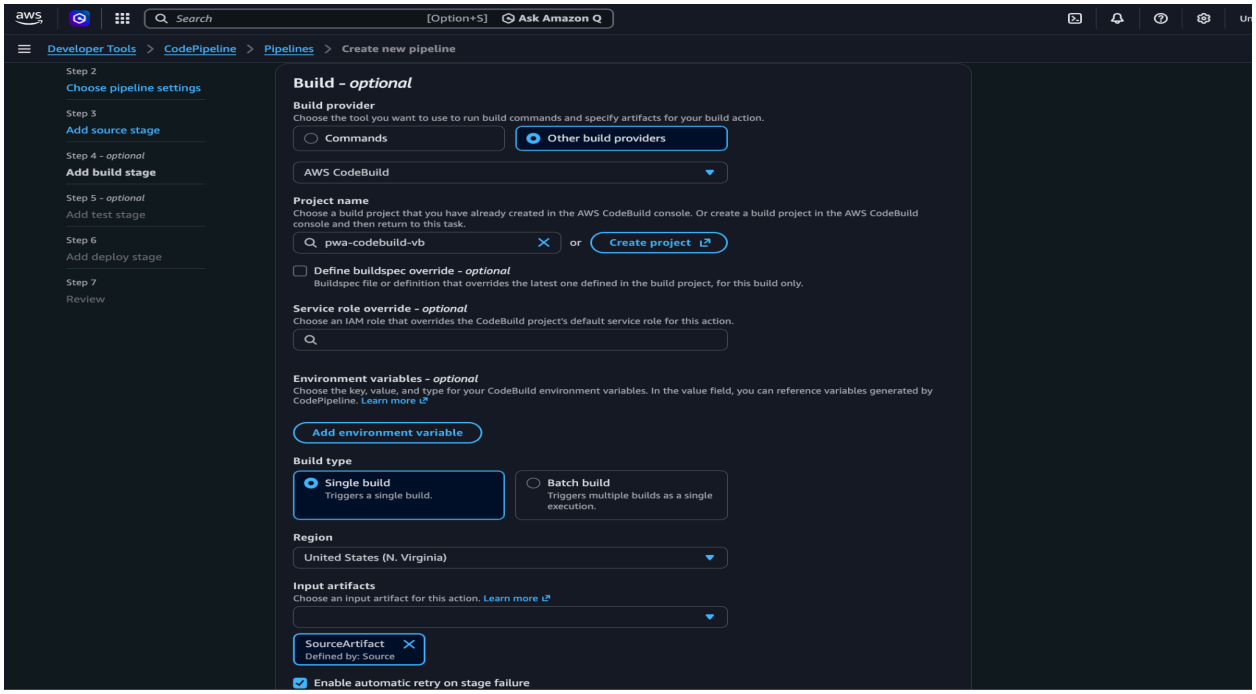
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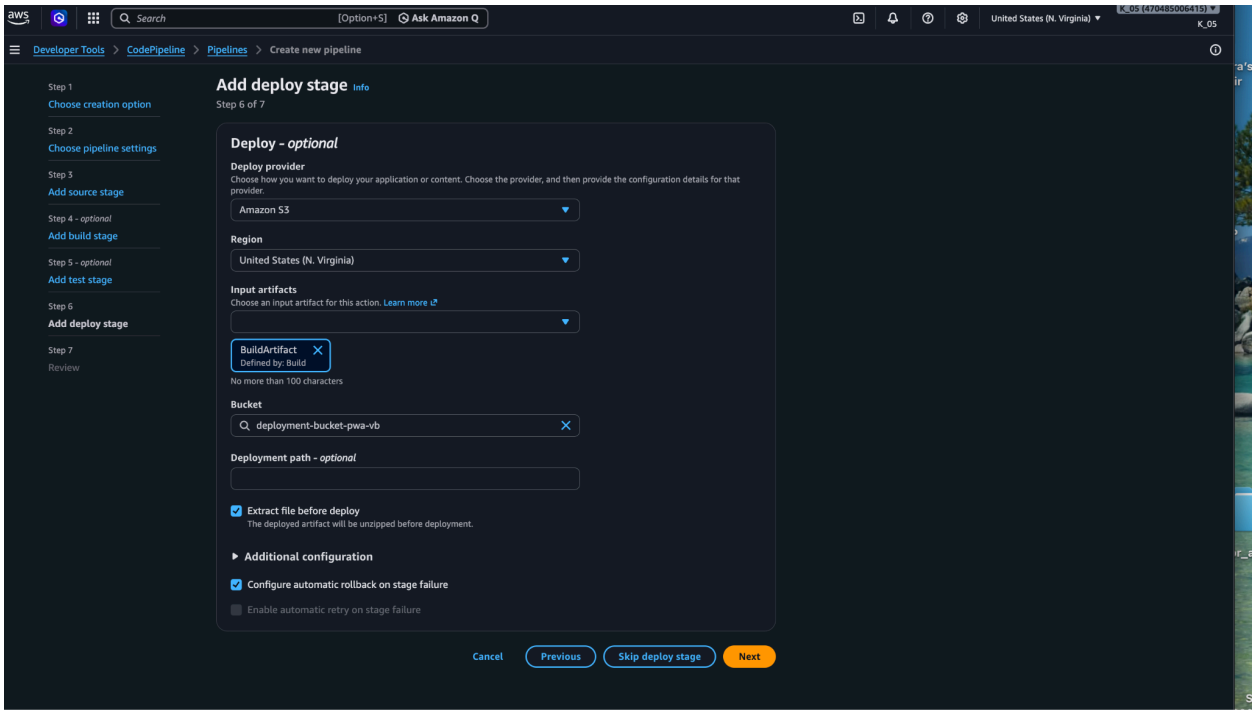


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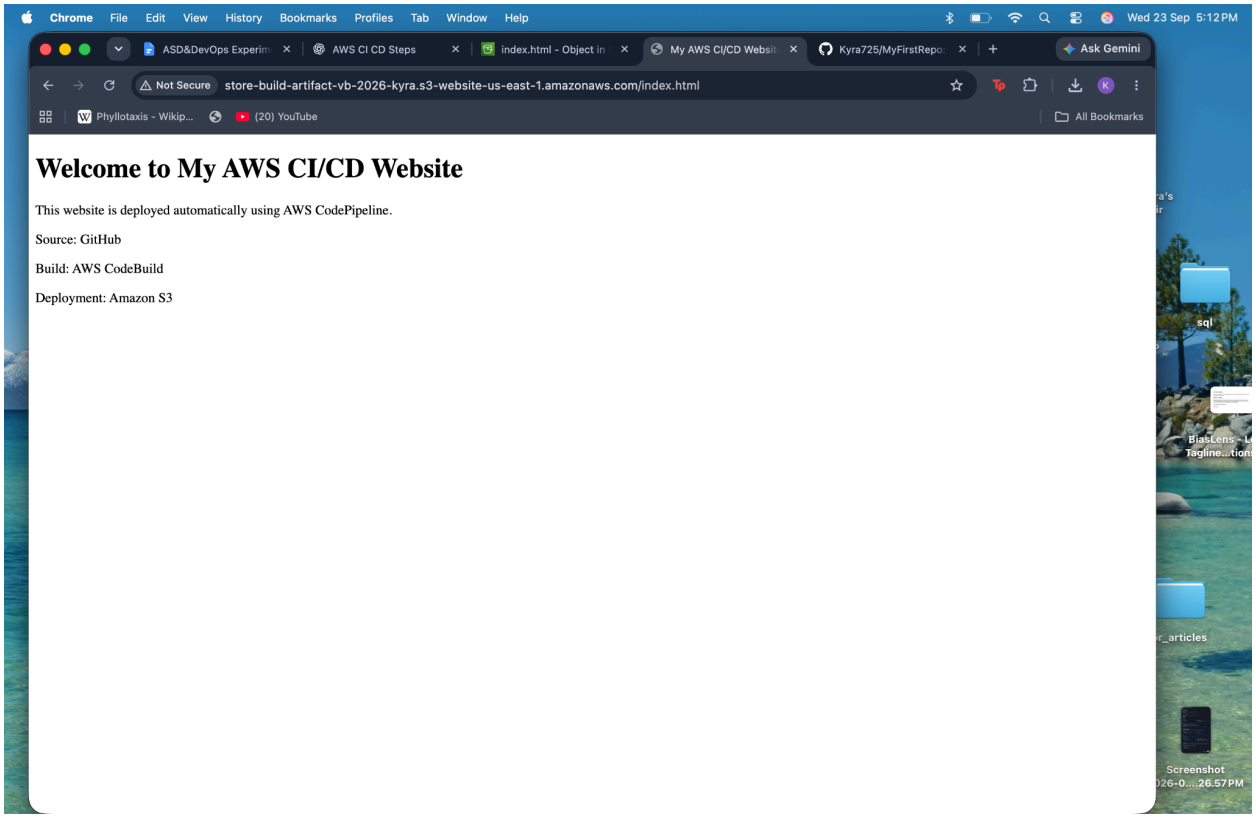


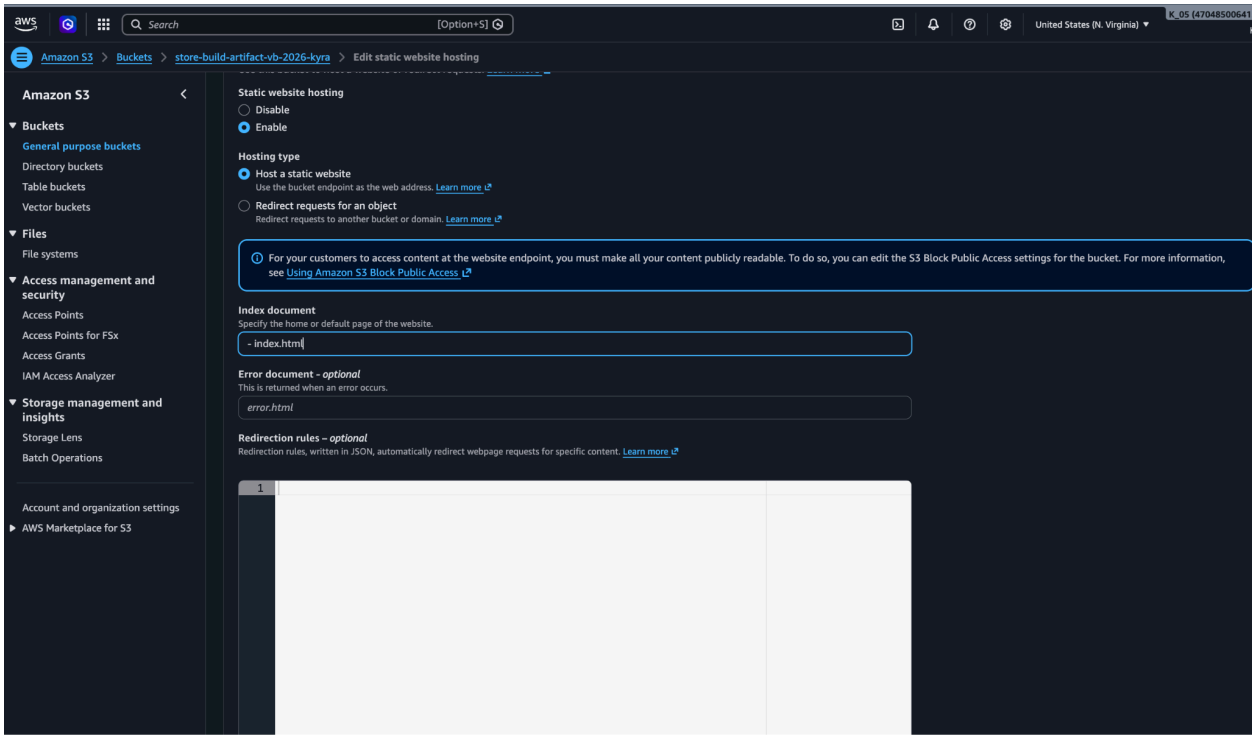
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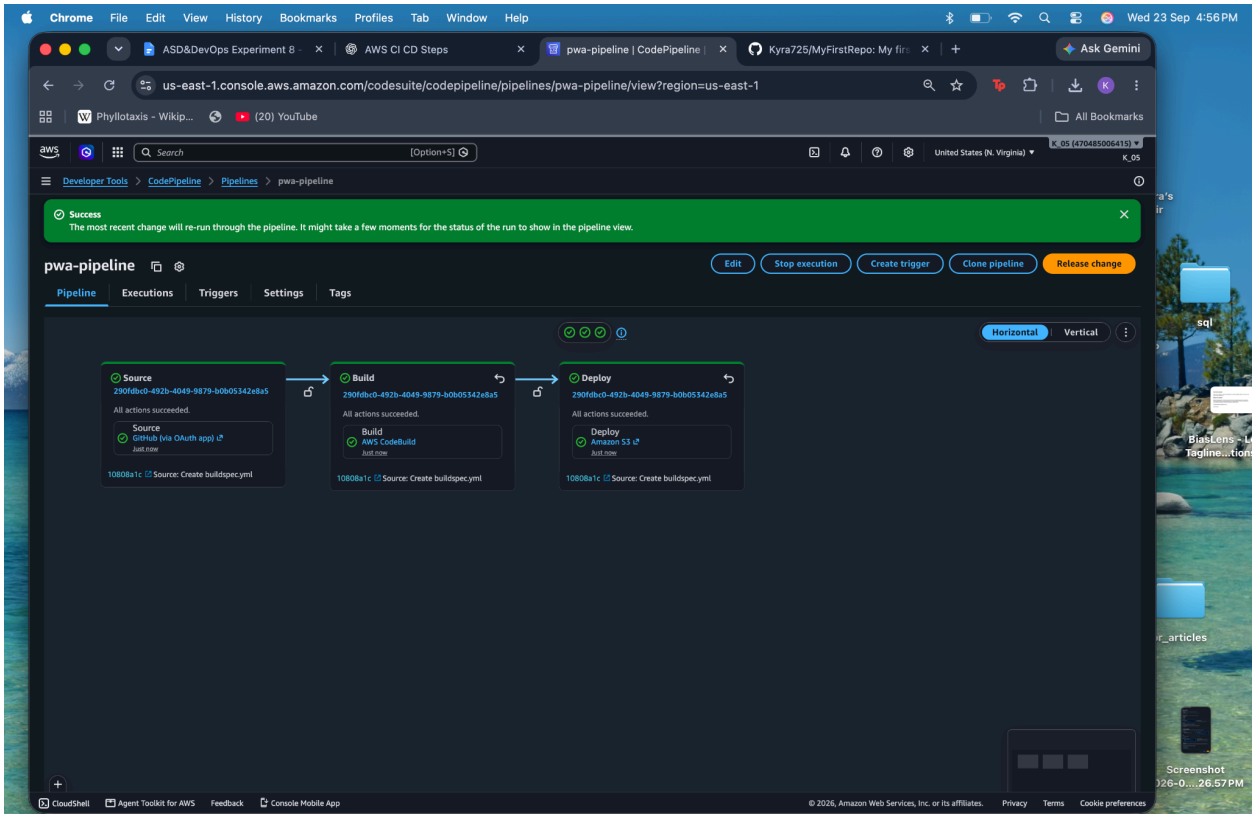


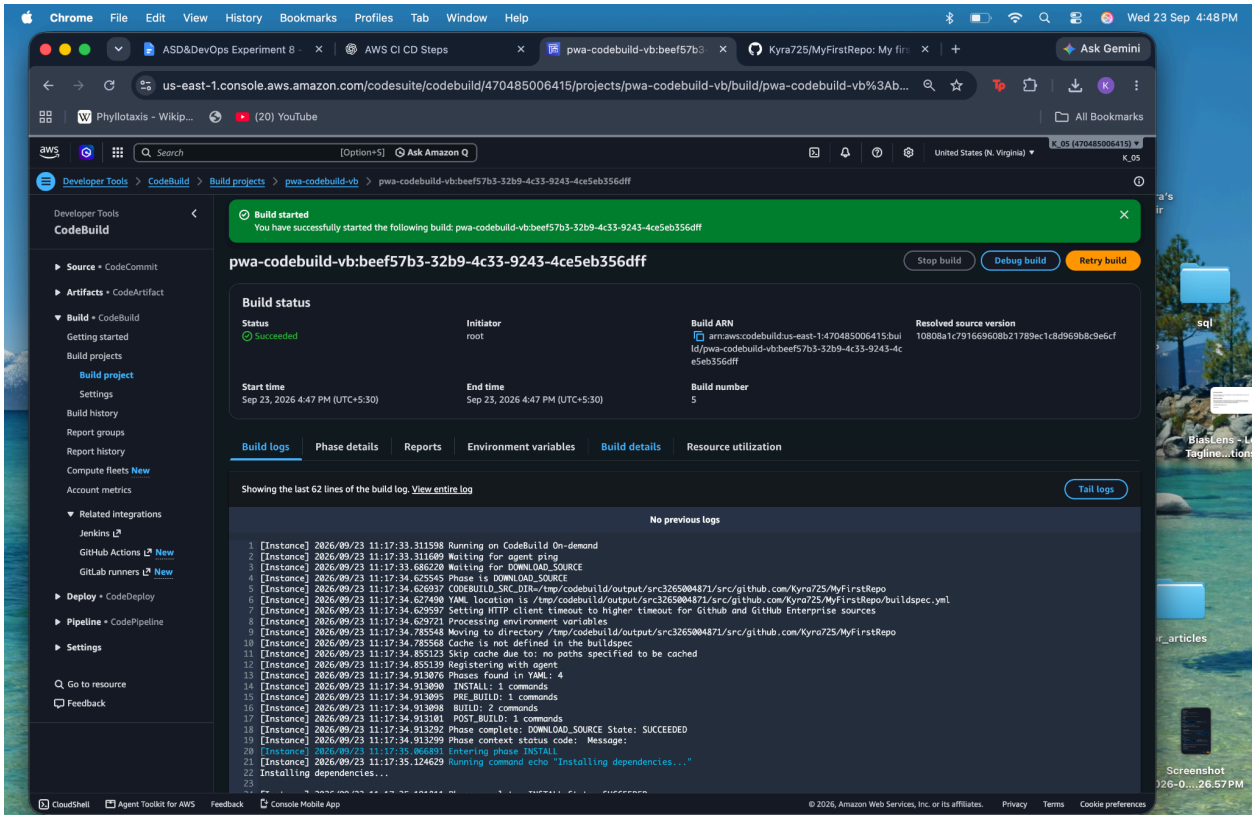
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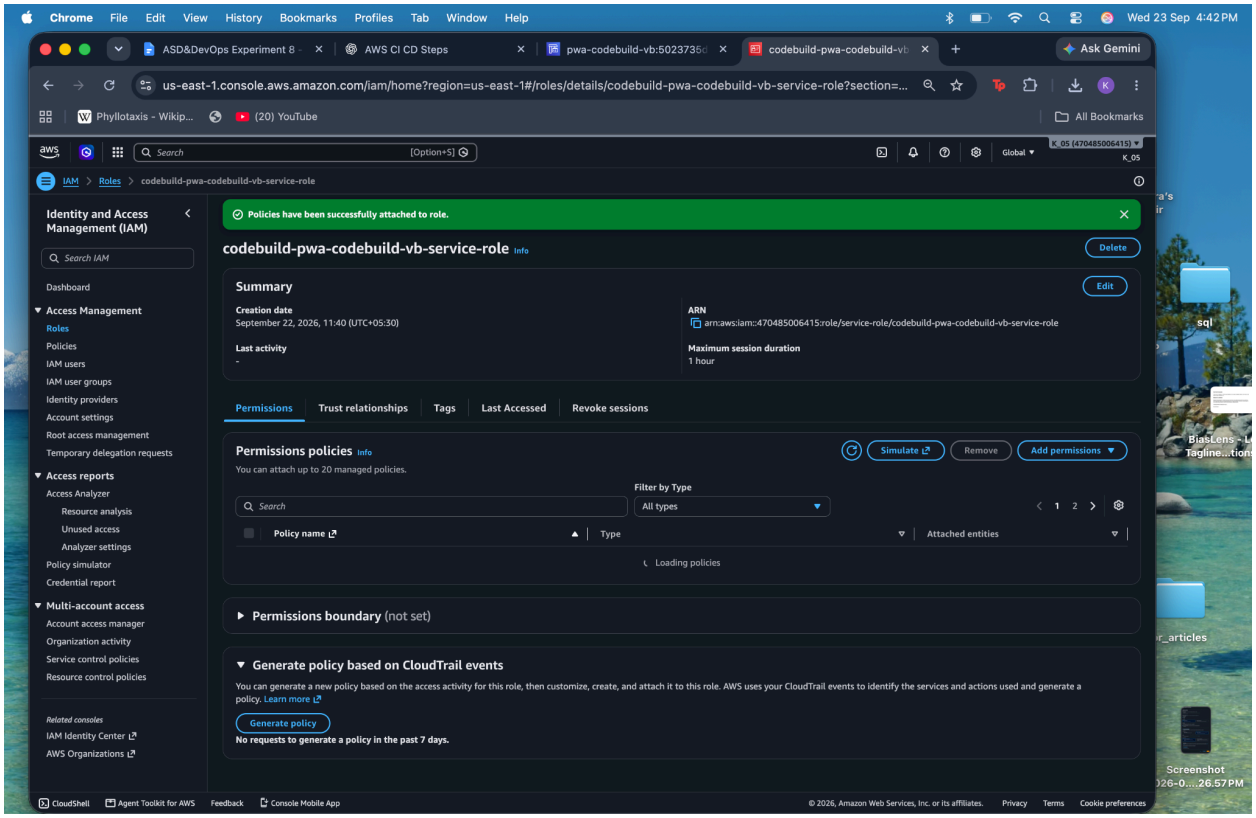


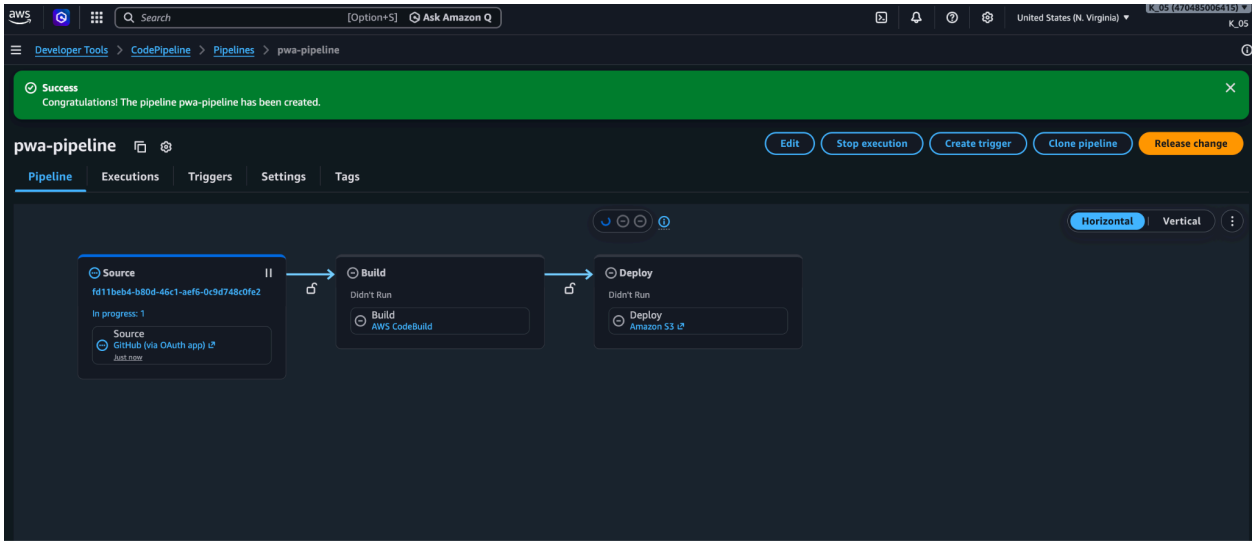
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Conclusion:

LO mapped successfully.